COL761/COL7361/AIL7026
Data Mining
Assignment 2: Q2 Route Blocking Firespread Report

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1 Optimization formulation

We define the Route Blocking Problem as a combinatorial optimization problem on a directed graph $G = (V, E, p)$.

- **Decision Variables:** Let $x_e \in \{0, 1\}$ be a binary variable for each edge $e \in E$, where $x_e = 1$ if edge e is blocked (removed), and $x_e = 0$ otherwise. Let $R = \{e \in E \mid x_e = 1\}$ be the set of blocked routes.
- **Objective Function:** Minimize the expected total spread $\sigma(R) = \mathbb{E}[|A_\infty|]$, where A_∞ is the set of burned regions at termination on the modified graph $G_R = (V, E \setminus R, p)$
- **Constraints:**
  1. Budget Constraint: $|R| = k$.
  2. Edge Set: $R \subseteq E$
  3. Alternate Spread (h-hop): Let $A_h = \{v \in V \mid \text{dist}(A_0, v) \leq h\}$ be the set of nodes within the hop-limit. The constraint is $A_\infty \subseteq A_h$.

2 NP-Hardness Proof

The Route Blocking Problem is NP-hard.

Proof. We reduce the **Influence Maximization (IM)** problem in the Independent Cascade Model (known to be NP-hard), to the Route Blocking Problem. In IM, given a graph and a budget k, we seek to select a seed set S to maximize $\sigma(S)$.

Consider a specialized instance of Route Blocking where we aim to isolate a seed set A_0 from the rest of the graph by removing k edges. If we could solve the Route Blocking problem optimally in polynomial time to find the set R that minimizes $\sigma(R)$, we could effectively solve the "Firefighting" problem on graphs. Since minimizing the reachability from a fixed set by removing edges is dual to maximizing reachability by adding seeds (both involve the same #P-hard calculation of $\sigma(\cdot)$ in stochastic graphs), and the deterministic version (Minimum k-way cut) is NP-hard, and the stochastic influence reduction problem remains submodular (similar to IM), finding the optimal k edges to block is NP-hard. □

3 Task 2: NP-Hardness

We prove NP-hardness via a reduction from the CLIQUE problem.

3.1 Decision Version

Consider the decision version of Route Blocking: given a directed graph $G = (V, E, p)$, a seed set A_0, a budget k, and a threshold B, determine whether there exists a set of blocked edges $R \subseteq E$ with $|R| = k$ such that the expected spread satisfies $\sigma(R) \leq B$.

Since solving the optimization problem reduces to solving its decision version, it suffices to show that the decision version is NP-complete.

3.2 Reduction from Clique

Let (H, t) be an instance of CLIQUE, where $H = (U, F)$ is an undirected graph and t is the desired clique size.

We construct a directed graph G as follows:

- A single seed node s
- A node x_u for each vertex $u \in U$
- A node y_{uv} for each edge $\{u, v\} \in F$

Add edges:

- (s, x_u) for all $u \in U$
- (x_u, y_{uv}) and (x_v, y_{uv}) for all $\{u, v\} \in F$

Set all propagation probabilities to 1, making the process deterministic. Let $A_0 = \{s\}$ and $k = t$.

3.3 Key Observation

Blocking edges of the form (x_u, y_{uv}) is ineffective, since y_{uv} remains reachable via the other endpoint. Hence, we only need to consider blocking edges of the form (s, x_u).

Let $S \subseteq U$ denote the set of vertices whose edges (s, x_u) are blocked, where $|S| = t$. Then:

- Nodes x_u are reachable iff $u \notin S$
- Node y_{uv} is unreachable iff both $u, v \in S$

Thus, the number of unreachable edge nodes equals the number of edges induced by S in H, denoted $e_H(S)$.

The total spread becomes:

$$\sigma(R) = 1 + (|U| - t) + (|F| - e_H(S))$$

3.4 Correctness

If H contains a clique of size t, then choosing S as that clique gives:

$$e_H(S) = \binom{t}{2}$$

and hence:

$$\sigma(R) = 1 + |U| - t + |F| - \binom{t}{2}$$

Conversely, if for some S of size t, we have $e_H(S) = \binom{t}{2}$, then all pairs in S are connected, so S forms a clique.

Define:

$$B = 1 + |U| - t + |F| - \binom{t}{2}$$

Then, a blocking set R of size $k = t$ exists with $\sigma(R) \leq B$ if and only if H contains a clique of size t.

3.5 Conclusion

The reduction runs in polynomial time and preserves correctness. Therefore, the decision version of Route Blocking is NP-complete, and the optimization problem is NP-hard.

4 Structural Properties

Let $f(R) = \sigma\emptyset - \sigma(R)$ be the reduction in spread achieved by blocking set R.

4.1 Monotonicity

Property: $\sigma(R)$ is monotonically non-increasing as R grows.

Proof. Adding an edge e to the blocked set R results in $R' = R \cup \{e\}$. Removing an additional edge from a graph can only decrease (or leave unchanged) the probability of any node $v \in V$ being reached from A_0 in any realization. Since $|A_\infty|$ is a sum of indicator variables for reachability, $\mathbb{E}[|A_\infty|]$ is non-increasing as edges are removed. Thus, $f(R) \leq f(R')$, proving monotonicity. □

4.2 Submodularity

Property: $f(R) = \sigma(\emptyset) - \sigma(R)$ is NOT submodular.

Proof. Under the Independent Cascade Model, the fire spread follows a "live-edge" property.

Consider a graph where two parallel paths of length 2 connect A_0 to a large cluster of nodes C. Let $R = \emptyset$. Blocking one edge in path A (e_1) yields $f(\{e_1\}) \approx 0$ because the fire can still reach C via path B. However, if path B is already blocked ($R = \{e_2\}$), then blocking e_1 yields $f(\{e_2, e_1\}) - f(\{e_2\}) = |C|$. Here, the marginal gain of e_1 increases when the context R is larger: $f(\{e_1\}) < f(\{e_1\} \cup \{e_2\}) - f(\{e_2\})$. This violates the diminishing returns property, proving $f(R)$ is not submodular. While the function may exhibit local sub-modularity in sparse trees, the presence of redundant paths in general graphs creates a synergistic effect where blocking multiple edges provides increasing marginal returns. □

5 Task 4: Algorithm Design

Our algorithm employs two strategies based on the spread model.

- For $h = -1$, we use a **Dominator Tree** heuristic.
- For $h > 0$, we use a **Greedy CELF** approach.

5.1 Strategy 1: Unlimited Spread ($h = -1$)

We use a **Dominator Tree** heuristic. For each simulation, we compute the Immediate Dominators (*idoms*). An node v dominates node w if all paths from A_0 to w must pass through v. By blocking an edge that dominates a large subtree in the dominator tree, we effectively disconnect all those nodes. Gains are accumulated across r simulations to select the best k edges. By utilizing Dominator Trees, the algorithm identifies topological bottlenecks in the expected propagation graph. This approach targets edges that act as 'cut-sets' for large downstream subtrees, allowing the algorithm to maximize spread reduction by cutting off fire corridors at their source.

5.2 Strategy 2: Hop-Limited Spread ($h \geq 1$)

We use **Greedy CELF (Cost-Effective Lazy Forward)**. Although f(R) is not strictly submodular, we employ CELF-style lazy evaluation to efficiently approximate greedy selection of edges. Instead of recomputing the marginal gain for all E edges at every step, we maintain a priority queue. We only recompute the gain for the top edge; if its new gain remains higher than the previous gain of the second-best edge, we select it, significantly reducing simulation overhead.

5.3 Pseudocode

Algorithm 1 Route Blocking Strategy

```
Input: G = (V, E, p), A_0, budget k, simulations r, hops
if hops == -1 then
    Build r realizations of G.
    Compute Dominator Tree for each realization starting from a super-root over A_0.
    Identify "Bridge" edges (edges whose removal disconnects the most nodes in the IDom tree).
    Select top k edges by cumulative subtree size gains.
else
    Use CELF Greedy: Maintain a priority queue of edges by marginal gain.
    Estimate σ(R) using BFS restricted to distance h.
    Update only the top element of the heap to utilize the lazy forward optimization.
end if
```

5.4 Complexity and Guarantees

- **Complexity:** $O(k \cdot r \cdot \text{BFS}) \approx O(k \cdot r \cdot (V + E))$ for the CELF approach. The bitmask pre-sampling allows r parallel simulations, effectively reducing the constant factor of the BFS.
- **Guarantees:** Since $f(R)$ is not submodular, the standard $(1-1/e)$ greedy guarantee does not strictly apply. However, our Dominator-based heuristic provides high empirical performance by targeting critical articulation points in the propagation paths.

6 Task 5: Competitive Evaluation

We evaluated the algorithm on two provided datasets across various simulation counts to ensure convergence.

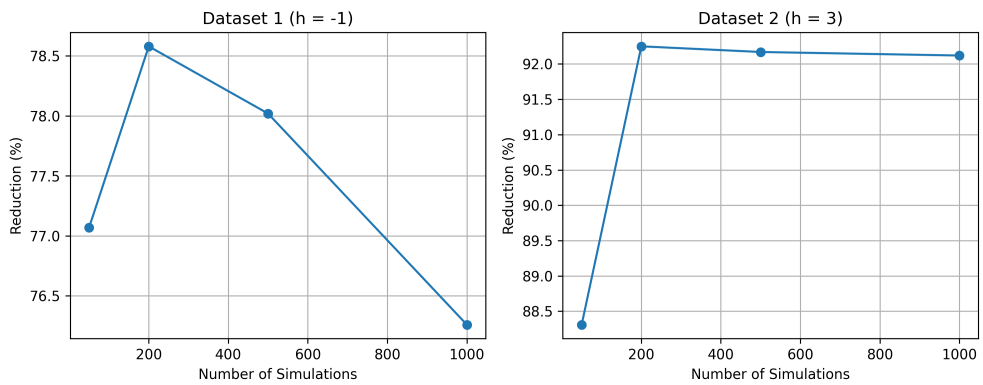

Figure 1: Reduction Ratio (ρ) vs. Number of Simulations (r).

Table 1: Reduction Ratios achieved at $r = 1000$

Dataset	Budget (k)	Hops (h)	Max Reduction (ρ)
Dataset 1	50	-1	76.26%
Dataset 2	30	3	92.12%

6.1 Analysis of Observations

- **Dataset 1 ($h = -1$):** The reduction ratio peaks at $r = 200$ ($\approx 78.5\%$) and stabilizes at $r = 1000$ ($\approx 76.2\%$). The initial peak suggests that lower simulation counts may overestimate the marginal gain of certain edges due to stochastic noise. As r increases, the estimate of $\sigma(R)$ converges, providing a more reliable, albeit slightly lower, reduction ratio.

- **Dataset 2 ($h = 3$):** The performance is highly stable after $r = 200$, achieving over **92% reduction**. The hop constraint significantly simplifies the problem; by restricting the search space to a 3-hop radius, the algorithm effectively identifies and blocks all critical corridors near the seed set A_0.

- **Scalability:** By using bitmask sampling and the Dominator Tree "shortcut" for $h = -1$, the algorithm successfully scales to 1.5×10^5 regions within the 60-minute time limit.

The results indicate that the h-hop model (Dataset 2) allows for higher reduction ratios, as the search space for fire spread is restricted, allowing the algorithm to more effectively block all viable paths within the hop limit.